

IT 462:
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Course Project
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Diabetes Health Indicators

Report

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1. Introduction

1.1. Background and Context

Diabetes is one of the most common chronic diseases worldwide and continues to create major pressure on healthcare systems. According to the World Health Organization (WHO), millions of individuals remain undiagnosed until serious complications begin to appear, including heart disease, kidney failure, and vision loss [1]. Early identification of high-risk individuals is important because timely intervention can reduce long-term medical complications and healthcare costs [1].

In recent years, healthcare organizations have increasingly relied on large-scale public health datasets to better understand the relationship between lifestyle behaviors and chronic diseases. The CDC Behavioral Risk Factor Surveillance System (BRFSS) is one of the largest health-related survey systems in the United States and contains information about physical activity, smoking habits, alcohol consumption, general health, and socioeconomic conditions [2]. Analyzing this type of data using big data technologies allows researchers to identify hidden patterns and trends that are difficult to detect using traditional methods.

This project focuses on the CDC Diabetes Health Indicators dataset obtained from the UCI Machine Learning Repository [3]. The dataset contains 253,680 survey responses with a mixture of binary, ordinal, and numerical attributes related to diabetes risk factors. Since the dataset contains a relatively large number of records and multiple feature types, Apache Spark was selected as the primary processing framework because of its scalability, distributed processing capabilities, and machine learning support.

1.2. Problem Statement

The main objective of this project is to analyze health and lifestyle factors associated with diabetes and to determine whether survey-based indicators can help identify individuals at higher risk of developing the disease. The project aims to explore relationships between variables such as BMI, physical activity, smoking behavior, mental health, and diabetes status using Spark RDD and SQL operations. In addition, machine learning techniques will be applied to build and evaluate a classification model capable of predicting diabetes risk based on health indicators. The results of this analysis may benefit healthcare professionals, public health organizations, and policymakers by supporting early detection efforts and improving preventive healthcare strategies.

1.3. Objectives

- To preprocess and clean the CDC Diabetes Health Indicators dataset using Apache Spark.
- To integrate, reduce, and transform the dataset into a consistent analysis-ready format.
- To explore important health and lifestyle patterns related to diabetes using Spark RDD and SQL operations.

- To build and evaluate a machine learning classification model for predicting diabetes risk.
- To interpret the analytical and machine learning results and communicate meaningful healthcare insights.

1.4. Scope and Limitation

1.4.1. Scope

This project is structured into multiple main phases: data preprocessing and RDD-based distributed analysis using Apache Spark. The goal is to prepare the CDC Diabetes Health Indicators dataset (BRFSS 2015) and then apply distributed computing techniques to extract meaningful health insights.

The implemented preprocessing tasks include:

- Detecting missing values and duplicate records
- Validating numerical ranges for selected health indicators
- Standardizing and validating the dataset schema
- Removing redundant attributes
- Creating engineered features such as BMI_Category and UnhealthyLifestyleScore

The RDD phase includes:

- Converting the dataset into structured RDDs (row-based and key-value pairs)
- Performing distributed aggregations using reduceByKey
- Applying filtering logic to identify high-risk individuals
- Computing statistical measures such as diabetes prevalence and average BMI
- Grouping data by demographic and health indicators (age, BMI category, health status)
- Sorting and ranking results to identify high-risk segments and population trends

All processed were performed only on the provided 2015 CDC dataset used in this project.

1.4.2. Limitations

The preprocessing phase has several limitations. The dataset is based on self-reported survey responses, which may contain inaccurate or inconsistent information. In addition, only the provided dataset was used, and no external healthcare or demographic datasets were integrated during preprocessing.

The preprocessing steps focused on basic data quality improvement and feature engineering. More advanced preprocessing techniques, such as complex imputation methods, automated feature selection, or large-scale optimization, were outside the scope of this phase.

Similarly, the RDD phase has its own limitations. While RDDs provide fine-grained control over distributed computation, they are more verbose and harder to maintain compared to DataFrame or SQL-based approaches. Performance tuning (e.g., partitioning and caching strategies) also requires manual configuration, and debugging distributed transformations can be more complex due to their functional execution model.

2. Data Selection and Understanding

2.1. Dataset Description

Source: The dataset used in this project is the CDC Diabetes Health Indicators dataset obtained from the UCI Machine Learning Repository

URL: <https://archive.ics.uci.edu/dataset/891/cdc+diabetes+health+indicators>

Domain: The dataset belongs to the healthcare and public health domain. It focuses on identifying relationships between lifestyle habits, general health conditions, and diabetes risk.

Size:

- Number of rows/records: 253,680
- Number of columns: 1 target variable + 21 original attributes
- Number of files/tables: 1 CSV file containing survey responses and health indicators
- Time coverage: BRFSS 2015 survey data

2.2. Structure and Schema

2.2.1. Data Schema

The dataset contains 21 attributes covering health indicators, lifestyle factors, and socioeconomic variables. The target variable is a three-class ordinal label reflecting diabetes status.

Variable Name	Role	Type	Attribute Group	Description	Example Values
Diabetes_binary	Target	Ordinary	Target Variable	Diabetes or prediabetes diagnosis	0 = No diabetes 1 = Prediabetes 2=Diabetes
HighBP	Feature	Binary	Health Indicator	Ever told you have high blood pressure	0 = No, 1 = Yes
HighChol	Feature	Binary	Health Indicator	Ever told you have high cholesterol	0 = No, 1 = Yes

CholCheck	Feature	Binary	Health Indicator	Cholesterol check in past 5 years	0 = No, 1 = Yes
Stroke	Feature	Binary	Health Indicator	Ever told you had a stroke	0 = No, 1 = Yes
HeartDiseaseorAttack	Feature	Binary	Health Indicator	Coronary heart disease or myocardial infarction	0 = No, 1 = Yes
DiffWalk	Feature	Binary	Health Indicator	Serious difficulty walking or climbing stairs	0 = No, 1 = Yes
Smoker	Feature	Binary	Lifestyle & Personal	Smoked at least 100 cigarettes in lifetime	0 = No, 1 = Yes
PhysActivity	Feature	Binary	Lifestyle & Personal	Physical activity in past 30 days (excl. job)	0 = No, 1 = Yes
Fruits	Feature	Binary	Lifestyle & Personal	Consume fruit 1 or more times per day	0 = No, 1 = Yes
Veggies	Feature	Binary	Lifestyle & Personal	Consume vegetables 1 or more times per day	0 = No, 1 = Yes
HvyAlcoholConsump	Feature	Binary	Lifestyle & Personal	Heavy drinker (men >14 drinks/wk; women >7 drinks/wk)	0 = No, 1 = Yes
Sex	Feature	Binary	Lifestyle & Personal	Biological sex	0 = Female, 1 = Male
AnyHealthcare	Feature	Binary	Social & Economic	Has any health care coverage (insurance, HMO, etc.)	0 = No, 1 = Yes
NoDocbcCost	Feature	Binary	Social & Economic	Could not see doctor in past 12 months due to cost	0 = No, 1 = Yes
GenHlth	Feature	Integer	Ordinal Scale	Self-rated general health status	1=Excellent → 5=Poor
Age	Feature	Integer	Ordinal Scale	13-level age category	1=18–24, 13=80+
Education	Feature	Integer	Ordinal Scale	Highest education level attained	1=No school → 6=College grad
Income	Feature	Integer	Ordinal Scale	Annual household income bracket	1=<\$10k → 8=≥\$75k
BMI	Feature	Integer	Continuous / Count	Body Mass Index	e.g., 25 (normal), 30 (obese)
MentHlth	Feature	Integer	Continuous / Count	Days mental health not good in past 30 days	Scale: 1–30 days

PhysHlth	Feature	Integer	Continuous / Count	Days physical health not good in past 30 days	Scale: 1–30 days
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Table 1. Dataset Key Features

2.3. Initial Data Quality Observations

Several observations were identified before formal preprocessing began.

- The dataset did not contain explicit null values.
- Duplicate survey records were identified and removed during preprocessing.
- Some numerical attributes contained values outside medically or logically valid ranges.
- BMI values showed the presence of extreme outliers and potentially incorrect entries.
- MentHlth and PhysHlth required range validation because valid responses should fall between 0 and 30 days.
- The dataset contained a mixture of binary, ordinal, and numerical variables, requiring schema standardization before analysis.

2.4. Ethical and Legal Considerations

The dataset is publicly available through the UCI Machine Learning Repository and originates from the CDC BRFSS survey program. The data is anonymized and does not contain personally identifiable information, making it appropriate for academic and educational use.

3. Data Preprocessing

3.1. Cleaning

The preprocessing phase started with data cleaning to improve the quality and reliability of the dataset. A full null-value scan was performed using Apache Spark to identify missing values across all attributes. No explicit null values were found in the dataset. Since the BRFSS survey uses numeric coding for responses, missing responses were not represented as SQL null values. Therefore, no imputation or missing-value replacement was required.

Duplicate survey responses were also checked using Spark's `dropDuplicates()` operation. A total of 23,899 duplicate records were identified and removed from the dataset. This reduced the number of rows from 253,680 to 229,781. Removing duplicate records helped ensure that each row represented a unique survey response and reduced the possibility of bias during analysis and model training.

Range validation was then applied to important numerical health indicators. BMI values were restricted to the range 10–80 based on medical plausibility and CDC BMI guidelines [2]. In addition, MentHlth and PhysHlth were limited to values between 0 and 30 because they represent the number of unhealthy days reported within the past month. Records outside these ranges were considered invalid and removed. After validation, the dataset size decreased from 229,781 to 229,500 records.

Issue	Before	After
Duplicate records	23,899 duplicates	0 duplicates
Invalid BMI / health values	281 invalid rows	0 invalid rows

3.2. Integration

Although the dataset consisted of a single CSV file, schema integration and standardization were still required before analysis. The target attribute **Diabetes_012** was renamed to **Diabetes** to improve readability and consistency throughout the project. The Diabetes column was also cast to Integer type to ensure compatibility with Spark SQL queries and machine learning tasks.

After integration and schema standardization, the dataset contained a mixture of integer and double attributes representing diabetes status, health conditions, lifestyle behaviors, and demographic indicators. The final schema included binary variables such as HighBP, Smoker, and PhysActivity, numerical variables such as BMI, MentHlth, and PhysHlth, and ordinal variables such as Age, Education, and Income. The target variable **Diabetes** was stored as an Integer type to support classification tasks in later phases of the project.

3.3. Reduction

Data reduction was applied to simplify the dataset and reduce unnecessary redundancy between features. The attribute **CholCheck** was removed from the dataset because it represents whether an individual checked their cholesterol level rather than an actual medical condition. Since the dataset already contains the **HighChol** feature, CholCheck was considered less informative and somewhat redundant.

The reduction process decreased the number of attributes from 22 to 21 while preserving the important health-related indicators needed for analysis.

Quantitative Impact

- Original rows: 253,680
- Final rows after preprocessing: 229,500
- Original attributes: 22

- Final attributes after reduction: 21

3.4. Transformation

Feature transformation and engineering were performed to improve interpretability and analytical usefulness. BMI values were transformed into categorical groups based on standard BMI classifications:

- 0: Underweight
- 1: Normal
- 2: Overweight
- 3: Obese

This transformation simplified the interpretation of BMI values and aligned them with commonly used medical categories.

An additional feature called **UnhealthyLifestyleScore** was also created using smoking status, alcohol consumption, and physical activity indicators:

$$\text{UnhealthyLifestyleScore} = \text{Smoker} + \text{HvyAlcoholConsump} + (1 - \text{PhysActivity})$$

The resulting score ranges from 0 to 3 and provides a simplified representation of unhealthy behavioral risk factors. This feature engineering step reduced reliance on multiple related binary variables while preserving important lifestyle information.

3.5. Summary of Preprocessing Pipeline

The preprocessing pipeline followed the sequence below:

Raw Dataset → Cleaning → Integration → Reduction → Transformation → Final Preprocessed Dataset

The preprocessing implementation was developed using Apache Spark and Scala. The corresponding code is stored in the project repository as:

[01_DataPreprocessing.scala](#)

4. [RDD Operations](#)

4.1. RDD Design

Data Model at RDD Level

In this project, the CDC Diabetes Health Indicators dataset was represented and processed using multiple RDD structures depending on the analysis task:

- **Raw parsed records:**

```
First record:  
0.0, 1.0, 1.0, 25.0, 1.0, 0.0, 1.0, 1.0, 0.0, 1.0, 0.0, 1.0, 0.0, 2.0, 0.0, 0.0, 1.0, 1.0, 11.0, 2.0, 3.0, 2.0, 1.0
```

- Each tuple represents a full row of 122 health attributes (e.g., BMI, Age, HighBP, Smoker, GenHlth, etc.).
- **Key-value transformations (most common structure):**
 - RDD[(Key, Value)]
 - Examples:
 - RDD[(AgeGroup, DiabetesFlag)]
 - RDD[(DiabetesStatus, BMI)]
 - RDD[(BMI_Category, 1)]
 - RDD[(RiskKey, 1)]
- **Aggregated structures:**
 - RDD[(Key, (Sum, Count))] for averages (e.g., BMI per diabetes status)
 - RDD[(Key, Rate)] for proportions (e.g., diabetes rate per age group)

Why RDDs were useful beyond DataFrames/SQL

RDDs were chosen in key parts of this analysis because they provide:

- **Fine-grained control over transformations**
 - Ability to explicitly control map, reduceByKey, and custom aggregation logic.
- **Flexible custom logic**
 - Easier implementation of multi-condition risk scoring (e.g., combined filtering of BP + smoking + inactivity).
- **Transparent computation steps**
 - Each transformation step is visible, improving interpretability for analytical validation.
- **Low-level optimization control**
 - More control over how intermediate data is structured compared to SQL abstractions.

4.2. Key Transformations and Actions

Analysis 1: Diabetes Rate by Age Group

Purpose:

To measure how diabetes prevalence changes across different age groups.

RDD Operations Used:

- **Transformations:** map, reduceByKey, mapValues, sortBy
- **Actions:** take, foreach

Pseudo-code:

```
val agePairs = dataRDD.map(r => (r.age, (r.diabetes, 1)))
val grouped = agePairs.reduceByKey((a,b) => (a._1 + b._1, a._2 + b._2))
val rates = grouped.mapValues{ case (d, total) => d.toDouble / total }
val sorted = rates.sortBy(_._2, ascending = false)
```

Sample Output:

2.3 Diabetes Distribution by Age Group			
Age Group	Diabetes Cases	Total	Diabetes Rate (%)
Age 1	177	5502	3.22%
Age 2	332	7057	4.70%
Age 3	697	9988	6.98%
Age 4	1390	12220	11.37%
Age 5	2256	14026	16.08%
Age 6	3787	17275	21.92%
Age 7	6556	23114	28.36%
Age 8	9012	27260	33.06%
Age 9	12060	29708	40.60%
Age 10	13639	29138	46.81%
Age 11	10775	22027	48.92%
Age 12	7204	15381	46.84%
Age 13	6850	16804	40.76%

Interpretation:

- Diabetes prevalence increases steadily with age.
- Older groups (60+) show significantly higher risk (>35–40%).
- Confirms age as a strong predictive risk factor.

Analysis 2: Average BMI by Diabetes Status

Purpose:

To determine whether BMI differs significantly between diabetic and non-diabetic individuals.

RDD Operations Used:

Transformations: map, reduceByKey, mapValues

Action: collect

Pseudo-code:

```
val bmiPairs = dataRDD.map(r => (r.diabetesStatus, (r.BMI, 1)))
val summed = bmiPairs.reduceByKey((a,b) => (a._1 + b._1, a._2 + b._2))
val avgBMI = summed.mapValues{ case (sum, count) => sum / count }
```

Sample Output:

2.4 Average BMI by Diabetes Status

Diabetes Status	Average BMI
No Diabetes	27.96
Prediabetes	30.68
Diabetes	31.90

Interpretation:

- Clear BMI gap exists between groups.
- Obesity strongly correlates with diabetes presence.
- Supports BMI as a major modifiable risk factor.

Analysis 3: High-Risk Population Identification

Purpose:

To identify individuals with multiple combined risk factors.

Conditions:

- HighBP = 1
- Smoker = 1
- PhysActivity = 0
- GenHlth ≥ 3

RDD Operations Used:

- **Transformation:** filter
- **Action:** count

Pseudo-code:

```
val highRisk = dataRDD.filter(r =>
  r.highBP == 1 &&
  r.smoker == 1 &&
  r.physActivity == 0 &&
  r.genHlth >= 3
)
```

```
val countRisk = highRisk.count()
```

Sample Output:

2.5 Identification of High-Risk Individuals

High-risk individuals count: 14313
Percentage of total population: 6.24%

Interpretation:

- A small but significant subgroup exhibits multiple risk factors.
- Risk factors frequently cluster rather than appearing independently.
- Highlights need for multi-factor intervention strategies.

Analysis 4: BMI Category Distribution

Purpose:

To understand population weight distribution and its link to chronic disease risk.

RDD Operations Used:

- **Transformations:** map, reduceByKey, sortBy
- **Actions:** take, foreach

Pseudo-code:

```
val categories = dataRDD.map(r => (bmiCategory(r.BMI), 1))
val counts = categories.reduceByKey(_ + _)
val sorted = counts.sortBy(_._2, ascending = false)
```

Sample Output:

2.6 BMI Category Distribution		
BMI Category	Count	Percentage
Obese	84661	36.89%
Overweight	82859	36.10%
Normal	58927	25.68%
Underweight	3053	1.33%

Interpretation:

- Over 70% of individuals are overweight or obese.
- Strong alignment with diabetes prevalence trends.
- Indicates population-level metabolic risk.

4.3. Key Transformations and Actions

Performance and Scalability Observations

- RDDs handled large-scale dataset (229,500+ records) efficiently due to Spark's distributed execution.
- reduceByKey significantly improved performance by minimizing data shuffling.
- Caching intermediate RDDs (where applied) reduced recomputation in iterative analyses (e.g., age and BMI groupings).

- Partitioning ensured parallel processing across transformations like map and filter.

Limitations and Difficulties

- **Verbose code structure**
 - Required multiple transformation steps for tasks that SQL/DataFrames could express in one line.
- **Debugging complexity**
 - Harder to trace lineage compared to DataFrame queries.
- **No schema enforcement**
 - Manual handling of data types increased risk of runtime errors.
- **Performance tuning burden**
 - Requires explicit optimization decisions (partitioning, caching).

Overall Evaluation

RDDs provided strong flexibility and control, making them ideal for custom analytical logic and multi-step transformations. However, they are less efficient in development speed and readability compared to higher-level Spark abstractions like DataFrames and SQL.

5. SQL Operations

5.1. DataFrame and View Creation

The preprocessed diabetes dataset was loaded into a Spark DataFrame using Spark SQL. The dataset was read from a CSV file with the header option enabled to recognize column names and the inferSchema option enabled to automatically detect data types.

The following code was used to load the dataset into a DataFrame:

```
scala> val diabetesDF = spark.read
      |   .option("header", "true")
      |   .option("inferSchema", "true")
      |   .csv("/Users/jood/Downloads/diabetes_preprocessed.csv")
val diabetesDF: org.apache.spark.sql.DataFrame = [Diabetes: int, HighBP: double ... 21 more fields]
```

After loading the dataset, the DataFrame was registered as a temporary SQL view named `diabetes_data` to allow SQL queries to be executed using Spark SQL.

```
scala> diabetesDF.show(5)
| diabetesDF.printSchema()
| diabetesDF.createOrReplaceTempView("diabetes_data")
```

5.2. Key Analytical Queries

Query 1: Diabetes Rate by Age Group

How does diabetes prevalence differ across age groups?

```
scala> spark.sql("""
  SELECT
    Age,
    COUNT(*) AS total_individuals,
    SUM(CASE WHEN Diabetes > 0 THEN 1 ELSE 0 END) AS diabetic_individuals,
    ROUND(
      SUM(CASE WHEN Diabetes > 0 THEN 1 ELSE 0 END) * 100.0 / COUNT(*),
      2
    ) AS diabetes_rate
  FROM diabetes_data
  GROUP BY Age
  ORDER BY Age
  """).show()
```

Interpretation:

The results show that diabetes prevalence generally increases as the Age code increases, rising from 1.80% in Age code 1 to a peak of 25.82% in Age code 11.

However, the rate decreases in Age codes 12 and 13.

Age is encoded as ordinal categories from 1 to 13, where higher values represent older age brackets, not literal ages. Therefore, values such as 11, 12, and 13 refer to older age brackets, not people aged 11 to 13.

Query 2: Average BMI by Diabetes Status

Do individuals with diabetes have higher BMI values compared to non-diabetic individuals?

```
scala> spark.sql("""
  SELECT
    Diabetes,
    COUNT(*) AS total_individuals,
    ROUND(AVG(BMI), 2) AS average_bmi
  FROM diabetes_data
  GROUP BY Diabetes
  ORDER BY Diabetes
  """).show()
```

Interpretation:

The results show that individuals with diabetes tend to have higher average BMI values compared to non-diabetic individuals. Non-diabetic individuals have an average BMI of 27.96, while individuals with prediabetes or diabetes have average BMI values above 30. This indicates a strong relationship between higher

body weight and diabetes risk.

Query 3: High-Risk Individuals Count

How many individuals have multiple major diabetes risk factors?

```
scala> spark.sql("""
  SELECT
    COUNT(*) AS high_risk_individuals
  FROM diabetes_data
  WHERE HighBP = 1
     AND Smoker = 1
     AND PhysActivity = 0
     AND Fruits = 0
     AND Veggies = 0
  """).show()
```

Interpretation:

The query identified 3,823 individuals with multiple diabetes-related risk factors, including high blood pressure, smoking, lack of physical activity, and low fruit and vegetable consumption. This suggests that unhealthy lifestyle habits and health conditions often occur together, which may significantly increase the risk of diabetes and other chronic diseases.

Query 4: Top Age Brackets by Diabetes Prevalence

Which age brackets have the highest diabetes prevalence?

```
scala> spark.sql("""
  SELECT
    Age,
    COUNT(*) AS total_individuals,
    SUM(CASE WHEN Diabetes > 0 THEN 1 ELSE 0 END) AS diabetic_individuals,
    ROUND(
      SUM(CASE WHEN Diabetes > 0 THEN 1 ELSE 0 END) * 100.0 / COUNT(*),
      2
    ) AS diabetes_rate
  FROM diabetes_data
  GROUP BY Age
  ORDER BY diabetes_rate DESC
  LIMIT 5
  """).show()
```

Interpretation:

The highest diabetes prevalence was observed in the older age brackets represented by Age codes 10 to 13, with the highest rate appearing in Age code 11 at 25.82%. These values represent ordinal age categories from the BRFSS dataset rather than literal ages. The results further support the strong relationship between increasing age and diabetes prevalence.

Query 5: BMI Category Distribution

How are individuals distributed across BMI categories?

```
scala> spark.sql("""
SELECT
  CASE
    WHEN BMI < 18.5 THEN 'Underweight'
    WHEN BMI >= 18.5 AND BMI < 25 THEN 'Normal'
    WHEN BMI >= 25 AND BMI < 30 THEN 'Overweight'
    ELSE 'Obese'
  END AS bmi_category,
  COUNT(*) AS total_individuals
FROM diabetes_data
GROUP BY
  CASE
    WHEN BMI < 18.5 THEN 'Underweight'
    WHEN BMI >= 18.5 AND BMI < 25 THEN 'Normal'
    WHEN BMI >= 25 AND BMI < 30 THEN 'Overweight'
    ELSE 'Obese'
  END
ORDER BY total_individuals DESC
""").show()
```

Interpretation:

The BMI category distribution shows that most individuals in the dataset fall within the overweight and obese categories. The obese category had the highest number of individuals at 84,661, followed closely by the overweight category at 82,859. This supports the earlier findings that elevated BMI is strongly associated with diabetes risk and highlights the prevalence of unhealthy weight ranges within the dataset.

5.3. Summary of Insights from SQL

- Diabetes prevalence increased significantly across higher age brackets, showing that age is a strong predictor of diabetes risk in the dataset.
- Individuals with diabetes had noticeably higher average BMI values compared to non-diabetic individuals, indicating a strong relationship between obesity and diabetes.
- A total of 3,823 individuals were identified with multiple combined risk factors such as high blood pressure, smoking, physical inactivity, and poor diet habits, suggesting that lifestyle-related features are important for prediction.
- The majority of individuals in the dataset were classified as overweight or obese, reinforcing the importance of BMI-related features in machine learning models.
- The SQL analysis revealed clear patterns and correlations between demographic and lifestyle factors and diabetes prevalence, which helped identify meaningful features for later machine learning tasks.

6. Machine Learning Operations

6.1. Problem Formulation

This phase focuses on building a machine learning model to predict diabetes status using health and lifestyle indicators from the CDC Diabetes Health Indicators dataset. The task is formulated as a multi-class classification problem because the target variable Diabetes contains three classes:

0 = No diabetes

1 = Prediabetes

2 = Diabetes

The input feature set includes medical, behavioral, and demographic indicators such as BMI, HighBP, HighChol, PhysActivity, Smoker, GenHlth, Age, Education, and Income. These variables were selected because they are commonly associated with diabetes risk and overall health condition.

One important constraint in the dataset is class imbalance. Most records belong to the non-diabetes class, which increases the baseline prediction accuracy and makes classification more challenging for minority classes.

6.2. Feature Pipeline

Apache Spark MLlib was used to build the machine learning pipeline. The pipeline included the following stages:

1. StringIndexer
The target variable Diabetes was converted into a numerical label column named label for compatibility with Spark ML classifiers.
2. VectorAssembler
Selected feature columns were combined into a single features vector used for model training.
3. RandomForestClassifier
A Random Forest classification model was trained using the assembled feature vectors.

The following features were included in the final model:

- HighBP
- HighChol
- BMI
- Smoker
- Stroke
- HeartDiseaseorAttack
- PhysActivity
- Fruits
- Veggies
- HvyAlcoholConsump
- AnyHealthcare
- NoDocbcCost
- GenHlth
- MentHlth
- PhysHlth
- DiffWalk

- Sex
- Age
- Education
- Income

These features were included because they represent important medical, behavioral, and socioeconomic indicators related to diabetes risk.

Pipeline Summary:

Raw Features → StringIndexer → VectorAssembler → RandomForestClassifier → Predictions

6.3. Model Choice

The Random Forest Classifier was selected for this project because it performs well on structured healthcare datasets containing both numerical and categorical-style variables. Random Forest models are also relatively robust to noise and nonlinear relationships between variables.

Another advantage of Random Forest is its ability to provide feature importance scores, which helps interpret which health indicators contribute most strongly to diabetes prediction.

The following hyperparameters were used:

Number of trees: 50
Maximum tree depth: 10
Random seed: 42

6.4. Training and Evaluation Setup

The dataset was divided using a 70/30 train-test split with a fixed random seed of 42 to ensure reproducibility.

Because the dataset is imbalanced toward the non-diabetes class, model performance was evaluated using both accuracy and F1-score rather than relying only on accuracy.

The following evaluation metrics were used:

- **Accuracy:** Measures the percentage of correctly classified records.
- **F1-Score:** Balances precision and recall and is more suitable for imbalanced classification problems.

A majority-class baseline classifier was also evaluated to compare whether the Random Forest model provided meaningful improvement beyond simply predicting the dominant class.

6.5. Results

Model	Accuracy	F1-Score	Baseline Accuracy
Random Forest Classifier	83.49%	78.28%	82.82%

The Random Forest model achieved slightly higher performance than the majority-class baseline classifier. However, the improvement was relatively small, indicating that class imbalance had a strong effect on prediction performance.

6.6. Interpretation

Feature importance analysis showed that general health condition, blood pressure, and BMI were among the strongest predictors of diabetes status.

6.6.1. Most Important Features

Feature	Importance
GenHlth	0.2101
HighBP	0.2065
BMI	0.1610
HighChol	0.0931
DiffWalk	0.0805

The results align with healthcare expectations because poor general health, obesity, and high blood pressure are widely recognized risk factors associated with diabetes.

The model performed well in identifying records belonging to the dominant non-diabetes class. However, predicting minority classes such as prediabetes remained more difficult because fewer examples were available during training.

6.6.2. Error Analysis

The prediction results showed that the model performed best on the majority non-diabetes class, while classification performance for prediabetes and diabetes cases was more limited. This behavior is mainly related to the imbalance in the dataset, where most survey responses belong to class 0 (no diabetes).

Some prediction errors may also be caused by similarities between classes. Individuals with prediabetes may share health characteristics with both healthy and diabetic participants, making separation between classes more difficult for the model.

Another possible source of error is the nature of the dataset itself. Since the BRFSS data is based on self-reported survey responses, some records may contain inaccurate or inconsistent information related to weight, physical activity, alcohol consumption, or general health condition.

In addition, the dataset does not include clinical measurements such as blood glucose levels, insulin resistance, or laboratory test results, which are important indicators for diabetes diagnosis. The absence of these features may reduce the model's ability to distinguish between similar health conditions accurately.

6.7. Limitations

Several limitations should be considered when interpreting the model results.

- The dataset is based on self-reported survey responses, which may contain inaccurate or biased information.
- The dataset does not include clinical measurements such as blood glucose levels or laboratory test results.
- The dataset is highly imbalanced toward the non-diabetes class, which increases baseline accuracy and limits the observed improvement of the machine learning model.
- No advanced hyperparameter tuning or class-balancing techniques were applied during this phase.
- Additional preprocessing, feature engineering, or alternative machine learning models may improve prediction performance in future work.

7. Conclusion and Future Work

7.1. Summary of Contributions

This project presented a full end-to-end big data analytics pipeline using the CDC Diabetes Health Indicators dataset (BRFSS 2015). The work covered multiple stages, including data preprocessing, distributed computation using Apache Spark RDDs, structured querying using Spark SQL, and machine learning-based pattern analysis.

Across these stages, the dataset was cleaned, transformed, and analyzed to extract meaningful public health insights. Feature engineering was applied to improve interpretability, and distributed processing was used to handle large-scale computations efficiently.

Key findings from the analysis include:

- **Strong age dependency:** Diabetes prevalence increases significantly with age, reaching over 40% in older age groups (80+).
- **BMI as a major risk factor:** Diabetic individuals consistently show higher average BMI compared to non-diabetic individuals.
- **Clustering of risk factors:** High blood pressure, smoking, inactivity, and poor health frequently co-occur in a measurable high-risk subgroup.
- **Population-level weight imbalance:** A large proportion of the dataset falls into overweight or obese categories, reinforcing obesity as a major public health concern.

These findings demonstrate that lifestyle and demographic factors can effectively predict and explain diabetes risk patterns in large-scale health datasets.

7.2. Reflection

This project provided practical experience in working with big data systems and understanding how different processing layers (RDDs, SQL, and ML models) interact within Apache Spark.

One of the key learnings was how **RDDs provide fine-grained control over distributed transformations**, while also requiring more verbose and manual implementation compared to higher-level abstractions like DataFrames and SQL. This highlighted the trade-off between flexibility and usability in big data frameworks.

Several practical challenges were encountered during implementation:

- **Performance considerations**, especially when handling large transformations and repeated actions
- **Debugging complexity** in RDD-based workflows due to lack of schema enforcement
- **Data quality issues**, including ensuring correct parsing and validation of raw survey data
- **Transformation design complexity**, particularly when combining multiple risk factors into a single analytical rule

Overall, the project reinforced the importance of choosing the right abstraction level depending on the task: RDDs for control, SQL for simplicity, and ML for predictive insight.

7.3. Future Work

Several improvements and extensions can be considered for future development of this project:

- **Additional data sources:**
Integrating external datasets (e.g., hospital records, socioeconomic data, or real-time health monitoring data) to improve prediction accuracy and contextual understanding.
- **Enhanced feature engineering:**
Expanding engineered features to include behavioral or temporal indicators, such as long-term risk scoring or weighted lifestyle indices.

- **Advanced machine learning models:**
Exploring more complex models such as Gradient Boosting, Random Forest tuning, or deep learning approaches, along with cross-validation and hyperparameter optimization.
- **Improved evaluation strategy:**
Using additional evaluation metrics such as ROC-AUC, precision-recall trade-offs, and calibration curves for better model interpretability in a medical context.
- **Deployment and integration:**
Packaging the trained model into a lightweight API or dashboard for real-time risk prediction, enabling integration with healthcare decision-support systems.

These enhancements would significantly increase the practical applicability of the system, moving it from an analytical prototype toward a deployable healthcare intelligence tool.

8. References

- [1] World Health Organization, *Global Report on Diabetes*, Geneva, Switzerland, 2016. [Online]. Available: <https://www.who.int/publications/i/item/9789241565257>
- [2] Centers for Disease Control and Prevention (CDC), “Behavioral Risk Factor Surveillance System (BRFSS),” 2015. [Online]. Available: <https://www.cdc.gov/brfss/>
- [3] CDC, “Diabetes Health Indicators Dataset,” UCI Machine Learning Repository, 2023. [Online]. Available: <https://archive.ics.uci.edu/dataset/891/cdc+diabetes+health+indicators>

9. Appendix

```
scala> diabetesDF.show(5)
+-----+
| diabetesDF.printSchema() |
| diabetesDF.createOrReplaceTempView("diabetes_data") |
+-----+

[Diabetes|HighBP|HighChol| BMI|Smoker|Stroke|HeartDiseaseorAttack|PhysActivity|Fruits|Veggies|HvyAlcoholConsump|AnyHealthcare|NoDocbcCost|GenHlth|MentHlth|PhysHlth|DiffWalk|Sex| Age|Education|Income|BMI_Cate
gory|UnhealthyLifestyleScore]

+-----+
| 0| 1.0| 1.0| 25.0| 1.0| 0.0| 1.0| 1.0| 0.0| 1.0| 0.0| 1.0| 0.0| 2.0| 0.0| 0.0| 1.0| 1.0| 11.0| 2.0| 3.0|
| 2| 0.0| 0.0| 28.0| 0.0| 0.0| 0.0| 1.0| 1.0| 1.0| 1.0| 0.0| 0.0| 2.0| 10.0| 0.0| 0.0| 1.0| 6.0| 5.0| 8.0|
| 0| 1.0| 1.0| 36.0| 1.0| 0.0| 0.0| 1.0| 1.0| 1.0| 0.0| 1.0| 0.0| 3.0| 0.0| 0.0| 0.0| 1.0| 11.0| 4.0| 7.0|
| 3| 1.0| 1.0| 23.0| 1.0| 1.0| 0.0| 1.0| 0.0| 0.0| 0.0| 1.0| 0.0| 3.0| 0.0| 2.0| 0.0| 1.0| 10.0| 4.0| 2.0|
| 1| 1.0| 1.0| 41.0| 1.0| 0.0| 0.0| 0.0| 0.0| 1.0| 0.0| 1.0| 0.0| 4.0| 3.0| 30.0| 1.0| 1.0| 9.0| 5.0| 6.0|
| 3| 2.0| 2.0| 2.0| 2.0| 2.0| 2.0| 2.0| 2.0| 2.0| 2.0| 2.0| 2.0| 2.0| 2.0| 2.0| 2.0| 2.0| 2.0| 2.0| 2.0|

only showing top 5 rows
root
 |-- Diabetes: integer (nullable = true)
 |-- HighBP: double (nullable = true)
 |-- HighChol: double (nullable = true)
 |-- BMI: double (nullable = true)
 |-- Smoker: double (nullable = true)
 |-- Stroke: double (nullable = true)
 |-- HeartDiseaseorAttack: double (nullable = true)
 |-- PhysActivity: double (nullable = true)
 |-- Fruits: double (nullable = true)
 |-- Veggies: double (nullable = true)
 |-- HvyAlcoholConsump: double (nullable = true)
 |-- AnyHealthcare: double (nullable = true)
 |-- NoDocbcCost: double (nullable = true)
 |-- GenHlth: double (nullable = true)
 |-- MentHlth: double (nullable = true)
 |-- PhysHlth: double (nullable = true)
 |-- DiffWalk: double (nullable = true)
 |-- Sex: double (nullable = true)
 |-- Age: double (nullable = true)
 |-- Education: double (nullable = true)
 |-- Income: double (nullable = true)
 |-- BMI_Category: integer (nullable = true)
 |-- UnhealthyLifestyleScore: double (nullable = true)
```

Figure 1, Data Frame creation and representation

```
(base) dalal@yousef@Dalal-MacBook-Air ~ % cd ~/Downloads/11442_Project
[spark-shell -i 01_DataPreprocessing.scala
WARNINGS: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
26/05/20 21:15:46 WARN Utils: Your hostname, Dalal-MacBook-Air.local, resolves to a loopback address: 127.0.0.1; using 192.168.68.100 instead (on interface en0)
26/05/20 21:15:46 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
Welcome to

      ____
     / ___/
    /  /_  /
   / ___/ /
  /  /_/_/
 /_____/

 version 4.1.1

Using Scala version 2.13.17 (OpenJDK 64-Bit Server VM, Java 21.0.10)
Type in expressions to have them evaluated.
Type :help for more information.
26/05/20 21:15:52 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Spark context Web UI available at http://192.168.68.100:4040
Spark context available as 'sc' (master = local[*], app id = local-17784369564323).
Spark session available as 'spark'.
(Diabetes,0)
(HighBP,0)
(HighChol,0)
(CholCheck,0)
(BMI,0)
(Smoker,0)
(Stroke,0)
(HeartDiseaseorAttack,0)
(PhysActivity,0)
(Fruits,0)
(Veggies,0)
(HvyAlcoholConsump,0)
(AnyHealthcare,0)
(NoDocbcCost,0)
(GenHlth,0)
(MentHlth,0)
(PhysHlth,0)
(DiffWalk,0)
(Sex,0)
(Age,0)
(Education,0)
(Income,0)
253680
229781
229781
229500
root
 |-- Diabetes: integer (nullable = true)
 |-- HighBP: double (nullable = true)
 |-- HighChol: double (nullable = true)
 |-- CholCheck: double (nullable = true)
 |-- BMI: double (nullable = true)
 |-- Smoker: double (nullable = true)
 |-- Stroke: double (nullable = true)
 |-- HeartDiseaseorAttack: double (nullable = true)
 |-- PhysActivity: double (nullable = true)
 |-- Fruits: double (nullable = true)
 |-- Veggies: double (nullable = true)
 |-- HvyAlcoholConsump: double (nullable = true)
 |-- AnyHealthcare: double (nullable = true)
 |-- NoDocbcCost: double (nullable = true)
 |-- GenHlth: double (nullable = true)
 |-- MentHlth: double (nullable = true)
 |-- PhysHlth: double (nullable = true)
 |-- DiffWalk: double (nullable = true)
 |-- Sex: double (nullable = true)
 |-- Age: double (nullable = true)
 |-- Education: double (nullable = true)
 |-- Income: double (nullable = true)
22
21
```

Figure 2, Data Preprocessing Output


```
scala> spark.sql("""
SELECT
  Age,
  COUNT(*) AS total_individuals,
  SUM(CASE WHEN Diabetes > 0 THEN 1 ELSE 0 END) AS diabetic_individuals,
  ROUND(
    SUM(CASE WHEN Diabetes > 0 THEN 1 ELSE 0 END) * 100.0 / COUNT(*),
    2
  ) AS diabetes_rate
FROM diabetes_data
GROUP BY Age
ORDER BY Age
""").show()
```

Age	total_individuals	diabetic_individuals	diabetes_rate
1.0	5502	99	1.80
2.0	7057	193	2.73
3.0	9988	384	3.84
4.0	12220	766	6.27
5.0	14026	1209	8.62
6.0	17275	2049	11.86
7.0	23114	3487	15.09
8.0	27260	4781	17.54
9.0	29708	6381	21.48
10.0	29138	7168	24.60
11.0	22027	5688	25.82
12.0	15381	3824	24.86
13.0	16804	3651	21.73

Figure 3, SQL Query 1

```
scala> spark.sql("""
SELECT
  Diabetes,
  COUNT(*) AS total_individuals,
  ROUND(AVG(BMI), 2) AS average_bmi
FROM diabetes_data
GROUP BY Diabetes
ORDER BY Diabetes
""").show()
```

Diabetes	total_individuals	average_bmi
0	189820	27.96
1	4625	30.68
2	35055	31.9

Figure 4, SQL Query 2


```
scala> spark.sql("""
  SELECT
    COUNT(*) AS high_risk_individuals
  FROM diabetes_data
  WHERE HighBP = 1
    AND Smoker = 1
    AND PhysActivity = 0
    AND Fruits = 0
    AND Veggies = 0
  """).show()
[
  +-----+
  |high_risk_individuals|
  +-----+
  |                    3823|
  +-----+
```

Figure 5, SQL Query 3

```
scala> spark.sql("""
  SELECT
    Age,
    COUNT(*) AS total_individuals,
    SUM(CASE WHEN Diabetes > 0 THEN 1 ELSE 0 END) AS diabetic_individuals,
    ROUND(
      SUM(CASE WHEN Diabetes > 0 THEN 1 ELSE 0 END) * 100.0 / COUNT(*),
      2
    ) AS diabetes_rate
  FROM diabetes_data
  GROUP BY Age
  ORDER BY diabetes_rate DESC
  LIMIT 5
  """).show()
[
  +-----+-----+-----+-----+
  |Age|total_individuals|diabetic_individuals|diabetes_rate|
  +-----+-----+-----+-----+
  |11.0|22027|5688|25.82|
  |12.0|15381|3824|24.86|
  |10.0|29138|7168|24.60|
  |13.0|16804|3651|21.73|
  |9.0|29708|6381|21.48|
  +-----+-----+-----+-----+
```

Figure 6, SQL Query 4

```
scala> spark.sql("""
  SELECT
    CASE
      WHEN BMI < 18.5 THEN 'Underweight'
      WHEN BMI >= 18.5 AND BMI < 25 THEN 'Normal'
      WHEN BMI >= 25 AND BMI < 30 THEN 'Overweight'
      ELSE 'Obese'
    END AS bmi_category,
    COUNT(*) AS total_individuals
  FROM diabetes_data
  GROUP BY
    CASE
      WHEN BMI < 18.5 THEN 'Underweight'
      WHEN BMI >= 18.5 AND BMI < 25 THEN 'Normal'
      WHEN BMI >= 25 AND BMI < 30 THEN 'Overweight'
      ELSE 'Obese'
    END
  ORDER BY total_individuals DESC
  """).show()
[
  +-----+-----+
  |bmi_category|total_individuals|
  +-----+-----+
  |Obese|84661|
  |Overweight|82859|
  |Normal|58927|
  |Underweight|3053|
  +-----+-----+
```

Figure 7, SQL Query 5

```

Last login: Sun May 10 20:32:11 on ttys000
(base) dalalalyousef@Dalals-MacBook-Air ~ % cd ~/Downloads/IT462_Project
[spark-shell -i 04_MachineLearning.scala
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
26/05/10 20:34:29 WARN Utils: Your hostname, Dalals-MacBook-Air.local, resolves to a loopback address: 127.0.0.1; using 192.168.68.100 instead (on interface en0)
26/05/10 20:34:29 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
Welcome to

  ____
 /  _ \
/_/_/  \_/_/  version 4.1.1

Using Scala version 2.13.17 (OpenJDK 64-Bit Server VM, Java 21.0.10)
Type in expressions to have them evaluated.
Type :help for more information.
26/05/10 20:34:37 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Spark context Web UI available at http://192.168.68.100:4040
Spark context available as 'sc' (master = local[*], app id = local-1778434479000).
Spark session available as 'spark'.
26/05/10 20:35:00 WARN SparkStringUtils: Truncated the string representation of a plan since it was too large. This behavior can be adjusted by setting 'spark.sql.debug.maxToStringFields'.
26/05/10 20:35:24 WARN DAGScheduler: Broadcasting large task binary with size 1072.8 KiB
26/05/10 20:35:28 WARN DAGScheduler: Broadcasting large task binary with size 2.0 MiB
26/05/10 20:35:34 WARN DAGScheduler: Broadcasting large task binary with size 3.8 MiB
26/05/10 20:35:39 WARN DAGScheduler: Broadcasting large task binary with size 1078.1 KiB
26/05/10 20:35:41 WARN DAGScheduler: Broadcasting large task binary with size 7.0 MiB
26/05/10 20:35:46 WARN DAGScheduler: Broadcasting large task binary with size 1861.6 KiB
26/05/10 20:35:48 WARN DAGScheduler: Broadcasting large task binary with size 3.6 MiB
26/05/10 20:35:52 WARN DAGScheduler: Broadcasting large task binary with size 3.6 MiB
Random Forest Model Performance
Accuracy = 0.8349282296650717
F1-Score = 0.7827613689845303
Baseline Accuracy = 0.828034076321625
Feature Importances
GenHlth -> 0.2101263108317552
HighBP -> 0.2065128870019166
BMI -> 0.16102559912807776
HighChol -> 0.09315212474624815
DiffWalk -> 0.08052757947188587
Age -> 0.0687640689665234
HeartDiseaseorAttack -> 0.04225804846674065
PhysHlth -> 0.03479981057616797
Income -> 0.02730458072089267
Education -> 0.013631126116692817
MentHlth -> 0.012603536477245238
HvyAlcoholConsump -> 0.009276788439923127
Stroke -> 0.007934568868120077
PhysActivity -> 0.0070871385120403795
Sex -> 0.006671546457058208
Smoker -> 0.004222655071859242
Veggies -> 0.0037331482953976663
NoDocbcCost -> 0.003634890742804805
Fruits -> 0.003454125190989905
AnyHealthcare -> 0.0032802879875313247
26/05/10 20:35:59 WARN DAGScheduler: Broadcasting large task binary with size 3.6 MiB
+-----+
|Diabetes|prediction|probability|
+-----+
|0|0.0|[0.9031408612935681,0.0813100563780553,0.015549082328376554]|
|0|0.0|[0.8752163249408691,0.10848849942475686,0.016295175634374061]|
|0|0.0|[0.8867378704772743,0.09642727271227021,0.016834856810455465]|
|0|0.0|[0.7948082434063455,0.1837210267635105,0.021470729830144114]|
|0|0.0|[0.8465192399460796,0.13805683310686712,0.015423926947053286]|
|0|0.0|[0.9789974969177564,0.016108236447849776,0.004894266634393844]|
|0|0.0|[0.9805915919552669,0.01466864339983661,0.004739764644896559]|
|0|0.0|[0.8979994988260622,0.08832857020423111,0.013671930967706649]|
|0|0.0|[0.9681124341872953,0.025957877586040654,0.005929688226664025]|
|0|0.0|[0.8843891029990582,0.1017363887886716,0.013874508212270377]|
+-----+
only showing top 10 rows

```

Figure 8, Machine Learning Output